

Saketh Annimalla

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

The University of Texas at Dallas , Richardson, TX	May 2025
Master of Science, Computer Science	GPA 4.00/4
Data structures & algorithms, Database design, Statistics for data science, Machine learning, NLP, Big data management	
Indian Institute of Technology Hyderabad	May 2021
Bachelor of Technology, Civil Engineering	GPA 8.17/10

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, JavaScript, TypeScript, C++, C
- **Frameworks:** FastAPI, Flask, Django, ReactJS, ExpressJS, Node JS, NextJS, Spring Boot, selenium, LangChain
- **Databases:** MySQL, PostgreSQL (pgvector, TimescaleDB), MongoDB, Redis cache, ClickHouse
- **Technologies:** Kafka, Kubernetes, Docker, Jenkins, gRPC, WebSocket, Grafana Loki, AWS S3, SageMaker, RDS, Lambda, EC2, JWT, Git, Jira, TailwindCSS, Hadoop, PySpark, MinIO, Superset, Airflow, Keycloak, LangFuse

PROFESSIONAL EXPERIENCE

BeeHyv Software Solutions , Dallas	May 2024 – January 2025
Software Developer Intern	
• Genie (RAG-based LLM Platform): • Developed chains using LangChain for a RAG based platform with configurable LLM abstraction for switching between multiple large language models like OpenAI GPT-4. • Built connectors for Google Drive and SharePoint to enable multi-source ingestion of various file types, utilizing Airflow, along with chunking and embedding, and integrated PostgreSQL with pgvector for semantic search and retrieval. • Integrated LangFuse for monitoring LangChain pipelines, evaluating model performance across different configurations. • Implemented Role-Based Access Control for multi-tenancy using Keycloak, integrated ClickHouse for event logging, and embedded Superset dashboards in ReactJS for visualizing ClickHouse data. • Tenxer Labs (Cloud-Based IC Evaluation Platform): • Developed the cloud controller and storage layers of a robust data pipeline using FastAPI, Kafka, gRPC and Time series PostgreSQL for cloud-based IC evaluation, processing real time data from remote labs providing accelerated IC adoption. • Transitioned inter service communication from WebSockets to gRPC following rigorous benchmark study achieving a 60% improvement in latency and throughput. • Created ReactJS components using TypeScript to display real time data visualization of IC evaluation results via WebSockets, utilizing compressed binary data reducing refresh latency by 2 times. • Implemented authentication system in React using React Query, Zustand featuring token-based API interception, role-based access control with JWT, automatic token refresh.	
University of Texas at Dallas , Richardson	December 2023 – April 2024
CS Outreach Instructor	
• Conducted hands on Java programming workshops with class sizes averaging 15 students to reinforce understanding of object oriented programming principles and problem solving techniques.	

BeeHyv Software Solutions , Hyderabad	July 2022 – July 2023
Software Developer	
• Built the ingestion module utilizing FastAPI to import large volumes of raw data in various formats from ERP systems and perform data preprocessing, making the data suitable for training models. Achieved a remarkable 95% reduction in processing time by implementing asynchronous batch insertions. • Defined and implemented data models using SQLAlchemy as ORM and Alembic for migrations, tailoring schemas for demand supply forecasting, inventory metrics, and analytics along with managing PostgreSQL. • Managed deployments, services, configuration of data pipelines by developing Kubernetes manifests, Helm charts.	
New Age IT , Mumbai	September 2021 – June 2022
Automation Engineer	
• Significantly improved workflow efficiency by configuring and managing Jenkins pipelines streamlining the software testing workflow through automation. • Achieved software reliability by developing comprehensive set of Python test scripts, rigorously validating various functionalities, including multi-source integrations and scalability, achieving software robustness. • Developed SQL query suites, assessing operator performance, memory utilization, and spillage occurrences, thus evaluating the software's efficiency.	

PROJECTS

- [E-commerce](#) - Developed a scalable e-commerce platform using a microservices architecture and **event-driven design** with Apache Kafka for inter-service communication. Utilized **Next.js**, **Nest.js**, PostgreSQL for Orders, **MongoDB** for Products, and Stripe API, with secure **OAuth 2.0** based authentication. Integrated Kong API Gateway, containerized with Docker and orchestrated on Kubernetes. Built a CI/CD pipeline using Jenkins and GitHub, with infrastructure automated by **Ansible** and monitoring setup with **Prometheus** and **Grafana**.
- [Portfolio-website](#) - Built a personal portfolio using ReactJS, Material UI, Chakra UI, and Tailwind CSS, showcasing projects and skills; deployed on GitHub Pages.
- [NewsAPI-ELK](#) - Built a real-time data pipeline using Kafka and PySpark Structured Streaming to process news data from NewsAPI, implemented Named Entity Recognition using NLTK , and visualized results with the ELK stack (Elasticsearch, Logstash, Kibana) for dynamic insights.
- [MeteororNN](#) - Developed a Recurrent Neural Network, along with Back propagation through time training algorithm, to predict Urban temperature using readings from multiple weather stations. Achieved a Root mean squared error of 10°F.
- Developed an Organic farming management system using Django for RestAPIs and ReactJS for intuitive UI, effectively processing & identifying fields from images using OpenCV, generating plans for a crop cycle and assigning tasks to users based on their roles.